Downloaded from geology.gsapubs.org on October 28, 2010

Geology

Early Tertiary paleomagnetic evidence and the displacement of southern Alaska

Gordon A. Thrupp and Robert S. Coe

Geology 1986;14;213-217
doi: 10.1130/0091-7613(1986)14<213:ETPEAT>2.0.CO;2

Email alerting services click www.gsapubs.org/cgi/alerts to receive free e-mail alerts when new articles cite this article

Subscribe click www.gsapubs.org/subscriptions/ to subscribe to Geology

Permission request click http://www.geosociety.org/pubs/copyrt.htm#gsa to contact GSA

Copyright not claimed on content prepared wholly by U.S. government employees within scope of their employment. Individual scientists are hereby granted permission, without fees or further requests to GSA, to use a single figure, a single table, and/or a brief paragraph of text in subsequent works and to make unlimited copies of items in GSA's journals for noncommercial use in classrooms to further education and science. This file may not be posted to any Web site, but authors may post the abstracts only of their articles on their own or their organization's Web site providing the posting includes a reference to the article's full citation. GSA provides this and other forums for the presentation of diverse opinions and positions by scientists worldwide, regardless of their race, citizenship, gender, religion, or political viewpoint. Opinions presented in this publication do not reflect official positions of the Society.

Notes

Geological Society of America

THE GEOLOGICAL SOCIETY OF AMERICA

Downloaded from geology.gsapubs.org on October 28, 2010

Early Tertiary paleomagnetic evidence and the displacement of southern Alaska

Gordon A. Thrupp, Robert S. Coe
Earth Sciences, University of California, Santa Cruz, California 95064

ABSTRACT

Paleomagnetic data from early Paleocene lavas, a few kilometres north of Lake Clark just inboard of the Peninsular terrane, indicate modest post-early Tertiary latitudinal displacement and about 55° of counterclockwise rotation of this region with respect to North America. A nearby middle Eocene sequence of lava flows, however, gives no indication of displacement or rotation. South of Lake Clark, the paleomagnetism of five Paleogene (30 to 40 Ma) volcanic sequences that are a part of the Peninsular terrane proper shows no northward translation and no rotation. A major part of the northward movement documented by other studies in the Gulf of Alaska region may be accounted for by cumulatively significant intraterrane displacements.

INTRODUCTION

The Peninsular terrane, composed chiefly of the Alaska Peninsula, is one of numerous lithotectonic terranes that make up Alaska (e.g., Silberling and Jones, 1984). A pioneering paleomagnetic investigation of Jurassic rocks on the Alaska Peninsula suggested substantial displacement of these rocks relative to the North American craton (Packer and Stone, 1974). Subsequent, more definitive paleomagnetic and paleontologic evidence concurs that in early Mesozoic time some of the southern Alaskan terranes were located thousands of kilometres to the south (e.g., Hillhouse, 1977; Newton, 1983).

Four paleomagnetic studies of Late Cretaceous and early Tertiary volcanic rocks (diamonds, Fig. 1) from terranes inboard of the Alaska Peninsula indicate no major latitudinal displacement (Hillhouse and Grommé, 1982; Hillhouse et al., 1985; Globerman and Coe, 1984; Coe et al., 1985). Hillhouse and Grommé's paleomagnetic study is of Paleocene lava flows inboard of Wrangellia near the bend in the surface trace of the Denali fault system (Fig. 1). Because the lavas are part of an apparently cogenetic Late Cretaceous–Paleocene magmatic belt (Hudson, 1979; Wallace and Engebretson, 1984) that intrudes the Peninsular terrane, Wrangellia, and neighboring inboard terranes, Hillhouse and Grommé suggested that the paleomagnetic result applies to the whole belt, which would imply that Wrangellia and the Peninsular terrane had arrived by Paleocene time. Subsequent results from Eocene volcanic rocks that lap onto the northern margin of Wrangellia (Fig. 1) corroborate this hypothesis (Hillhouse and Grommé, 1984).

In contrast, paleomagnetic data from several studies of Upper Cretaceous and Tertiary rocks from outboard parts of the Alaska Peninsula (solid circles, Fig. 1) have been interpreted as requiring thousands of kilometres of northward displacement of the Peninsular terrane throughout Tertiary time (Stone et al., 1982). The displacements indicated, however, are highly variable and in some cases internally inconsistent. Because the data are all from detrital sedimentary rocks, the apparent displacements may be, in part, artifacts of inclination error, a spurious shallowing of inclination exhibited by some clastic sedimentary rocks (e.g., Coe et al., 1985). Furthermore, it is probable that some of the low paleolatitudes are suspect because they may have been derived from unrecognized magnetic overprints that occurred after folding (Coe et al., 1985). Plutonism and volcanism have occurred throughout much of the Tertiary on the Alaska Peninsula, and for most of these studies there is no demonstration that secondary components of magnetization have been completely removed.

Outboard of the Peninsular terrane, however, low paleolatitudes inferred from more reliable paleomagnetic data from igneous rocks (triangles, Fig. 1) are difficult to refute; relative to North America, the mean results suggest about 25° of northward movement of both the Prince William and the Chugach terranes during Tertiary time (Plumley et al., 1983; Grommé and Hillhouse, 1981). The marked discrepancy between the early Tertiary paleolatitudes for terranes inboard and outboard of the Peninsular terrane indicates juxtaposition of major parts of southern Alaska during Tertiary time—but an appropriate tectonic boundary is not known.

To investigate the late Mesozoic–Cenozoic development of southern Alaska, we collected samples of Late Cretaceous–Paleogene volcanic rocks from the Alaska Peninsula for paleomagnetic analysis. In this paper our purpose is to summarize the results and present new constraints on the tectonic evolution of southern Alaska.

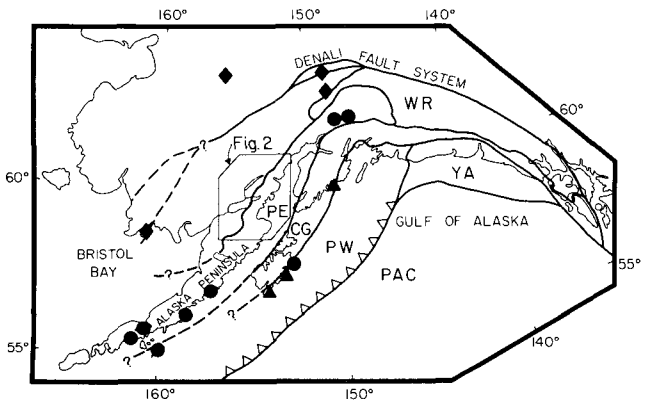


Figure 1. Map of southern Alaska showing lithotectonic terranes after Silberling and Jones (1984), and Late Cretaceous–early Tertiary paleomagnetic study localities (see text for references). Terranes: WR = Wrangellia-Alexander-Craig composite; PE = Peninsular; CG = Chugach; PW = Prince William; PAC = Pacific plate. Paleomagnetic studies: diamonds—no major northward displacement (volcanic rocks); solid circles—major northward displacement (sedimentary rocks); triangles—major northward displacement (igneous rocks).

Downloaded from geology.gsapubs.org on October 28, 2010

EXPERIMENTAL PROCEDURES AND ASSESSMENT OF THE PALEOMAGNETIC DATA

The paleomagnetic data discussed here are from seven K-Ar dated localities, a total of 87 flows. Each locality is a sequence of superposed basaltic or andesitic, subaerial lava flows where the division between flows and the tilt of the sequence are discernible. In most of the volcanic sequences, ashy zones or soils occur between flows. In a few cases, however, oxidized, vesicular flow tops or brecciated bases are the only indications of divisions between flows. Four to ten oriented cores were collected from each lava flow.

Detailed pilot thermal and alternating field (AF) incremental demagnetization experiments (about 15 steps) were done on different segments of a single core from every lava flow. The better method of cleaning was usually thermal, less frequently AF, or AF followed by thermal when lightning strikes were suspected. This was applied in 3–6 steps to the other samples of the flow. For every sample, a least-squares algorithm was used to determine the best-fit cleaned direction from demagnetization steps chosen with the aid of a vector diagram.

Very little variation of attitude occurs within or between the various localities of similar age; therefore, significant fold tests are not possible. Nonetheless, there is strong evidence that stable, single-component directions of magnetization have been isolated that are almost certainly primary: very low-grade alteration observed in thin sections, low blocking temperature of secondary components, well-defined characteristic components, and nearly antipodal normal and reversed mean directions.

Although considerable care and effort were spent obtaining attitudes, true paleohorizontal is a potentially significant uncertainty for each locality. Results from tilted sequences were corrected on the assumption that the flows were initially horizontal, but some of the observed tilt is undoubtedly initial dip of the flows. However, for each of the sequences sampled, the nearly uniform thickness of flows and tilt of extensive exposures suggest that initial dips were probably gentle. The five Iliamna quadrangle localities are spaced over a large region (10 000 km^2) with numerous early Tertiary volcanic vents (Detterman and Reed, 1980). Consequently, it is likely that most initial-dip error is averaged out when the data are considered collectively.

Because volcanism is sporadic, an estimate of the mean ambient field direction during the time spanned by a sequence of lava flows is usually improved if results from adjacent flows that have very similar directions are combined. Failure to combine data or to use an appropriate weighting scheme results in a confidence limit that is too low and a mean that is biased toward the direction of the field when flows were most rapidly extruded (Cox and Gordon, 1984). We evaluated the data from each of the seven sections and combined sequential flows when the magnetic directions and field evidence suggested that a group of flows spanned very little time. We present the paleomagnetic results obtained by using both the preferred, grouped-flows method and the more conventional practice of weighting each flow equally.

EARLY PALEOCENE RESULTS FROM NORTH OF LAKE CLARK

The early Paleocene (66 Ma) locality (see Fig. 2) consists of two stacks of basaltic lava flows that are about 4 km apart; collectively, these comprise a 200-m sequence of at least 38 distinct flows, 30 of which were sampled. The magnetic polarity stratigraphy (Fig. 3a) and the nearly antipodal means of normal and reversed flows (Fig. 3d) demonstrate that secular variation is averaged adequately, thereby allowing legitimate inference of paleolatitude. The whole-rock K-Ar age on one flow in this sequence (65.8 ± 13 Ma) has a large uncertainty but falls within the range (59–71 Ma) of ten other K-Ar ages (Eakins et al., 1978) on plutons, an andesitic flow, and dacitic domes in this region of the Lake Clark quadrangle. We refer to this locality as early Paleocene in age.

A mean is calculated both (1) by giving each flow equal weight (N [flows] = 30) and (2) by combining results from adjacent flows that appear to be temporally grouped (N [groups] = 13). Individual flow and flow-group data are depicted in Figures 3b and 3c. Both means suggest approximately 9° of northward displacement and 55° of counterclockwise rotation with respect to North America since the early Paleocene (see also bottom of Table 1). If the section is younger than 65 Ma, then because of the southward motion of North America since the early Tertiary, the indicated northward movement would be even less.

At face value the flow-group–data mean indicates 8.7° of northward displacement. Although northward displacement in this case is not significant at 95% confidence (±10.7°), it is significant at 85% (±8.2°), and of course is even more so at 63% (±5.8°). Thus, it is likely that at least a modest amount (several hundred kilometres) of northward movement has occurred, but major displacement (20° or more) is unlikely. On the other hand, the paleomagnetic data from the Paleocene Ghost Rocks Formation on Kodiak Island (Plumley et al., 1983) indicate, with a high degree of confidence, major northward movement (25° ±7°) of the Prince William terrane.

EOCENE—OLIGOCENE PALEOMAGNETIC RESULTS FROM SOUTH OF LAKE CLARK

South of Lake Clark, we sampled a total of 50 flows from five sequences of superposed andesitic lava flows and several isolated groups of only a few flows. These rocks range in K-Ar age from 30 to 40 Ma and are distributed over a major part of the Iliamna quadrangle (see Fig. 2). They are inboard of, but locally lap onto, the Jurassic plutonic spine of the Alaska Peninsula (Detterman and Reed, 1980) and hence are a part of the Peninsular terrane as defined by Jones et al. (1981).

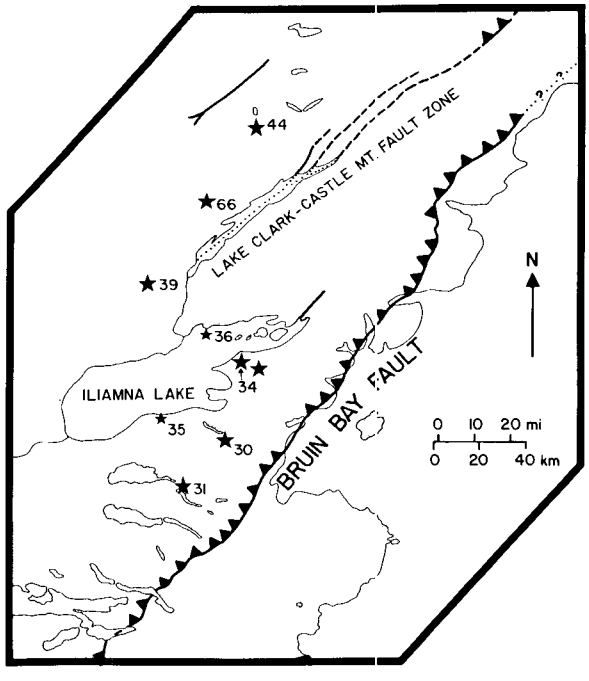


Figure 2. Map of Lake Clark–Iliamna Lake region of Alaska Peninsula. Stars mark sampled localities (each a sequence of lava flows); numbers indicate K-Ar ages.

Downloaded from geology.gsapubs.org on October 28, 2010

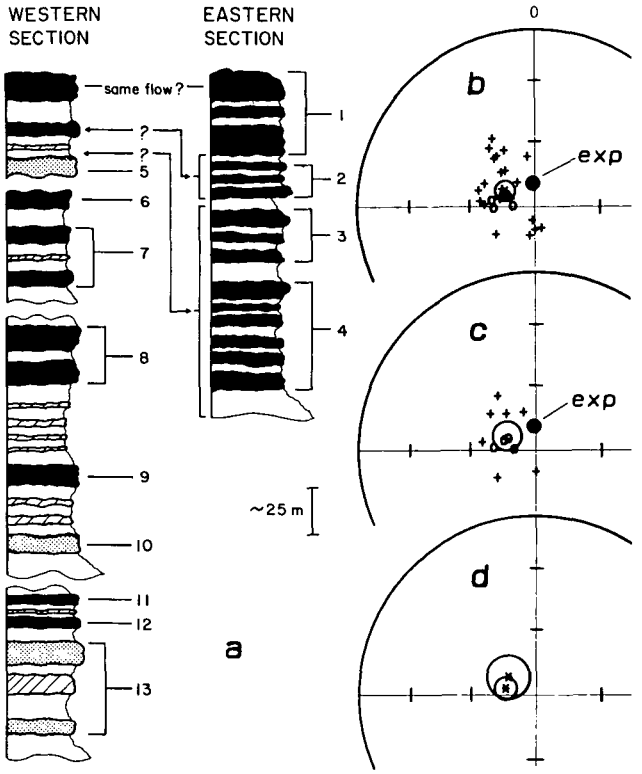


Figure 3. Paleomagnetic results from 30 lower Paleocene basaltic lava flows north of Lake Clark. a: Columnar section showing magnetic polarity stratigraphy and flow grouping (numbers). Black—normal-polarity flows; gray (stippled)—reversed-polarity flows; diagonal rule—unsampled flows; blank (white) space between flows—soil, ash, tuff, or covered section. b: Stereographic equal-area projection on lower hemisphere of flow-mean directions and their mean enclosed by its circle of 95% confidence (alpha-95). Reversed (o) flow mean directions are inverted for comparison with normal (+) ones (same for c and d). Large solid circle "exp" is alpha-95 about expected direction for locality if it had been in its present location with respect to North America when lavas erupted (same for Fig. 4). c: Flow-group mean directions, their mean and alpha-95. d: Reversal test: mean of four reversed flow-group means does not differ significantly from mean of nine normal flow-group means. Normal mean has larger alpha-95.

TABLE 1. SUMMARY OF PALEOCENE PALEOMAGNETIC DATA FROM NORTH OF LAKE CLARK

	N/N$_0$	Uncorrected for tilt: Dec$^\circ$	Inc$^\circ$	a95°	k	R	SD$^\circ$	Corrected for tilt: Dec$^\circ$	Inc$^\circ$	a95°	k	R	SD$^\circ$	Mean Virtual Geomagnetic Pole: Plong$^\circ$	Plat$^\circ$	A95°	K	R	SD$^\circ$	Colat$^\circ$
Paleocene Locality*																				
Method I (each flow given equal weight):	30/30	337.9	75.4	4.8	31.6	29.0813	14.4	298.6	74.4	4.7	32.8	29.1146	14.2	146.8	60.7	7.9	12.0	27.5742	23.4	27.9
Method II (flows grouped):	13/13	333.6	77.9	6.8	37.7	12.6820	13.2	296.1	75.2	6.7	38.0	12.6904	13.0	149.3	60.5	11.7	13.5	12.1116	22.0	26.8

* Locality location: 205.34°E 60.27°N. Attitudes: Section 1 (flows 1-16) N 45°W strike, 13°SE dip. Section 2 (flows 17-30) N 20°W strike, 8°E dip. Each attitude is constructed from apparent dips shot from a distance. Relative to a 65 Ma pole for North America, 77.8°N, 190.9°E, A95 = 7° (interpolated; see Coe et al., (1985)), the mean of the VGPs suggests 9.7° ± 8.3° (Method I) or 8.7° ± 10.7° (Method II) of northward displacement and 53.5° ± 22.6° (Method I) or 55.0° ± 27.8° (Method II) of counterclockwise rotation of this part of Alaska since Early Paleocene time. The 95% confidence limits on displacement and rotation are estimated using bivariate to univariate adjustment factors Demarest (1983).

The Peninsular terrane paleomagnetic data are summarized in three ways: all flows averaged (Fig. 4a), flow groups averaged (Fig. 4b), and flow-group–locality means averaged (Fig. 4c). In each case, the mean direction of the Peninsular terrane data is nearly identical to the direction expected if this region had been fixed to North America since the lavas were extruded. At each locality except one the flows are all one polarity. We are not confident that any one of these lava-flow piles records enough range of secular variation to stand alone as an indicator of paleolatitude. Collectively, however, there are 22 normal and 28 reversed flows, and the 95% confidence circles about the means of the two polarity groups overlap. The slight departure from antiparallel is probably due to imperfect averaging of secular variation. There is no significant fold test because the sequences are nearly flat-lying.

DISCUSSION

The paleomagnetism of the early Paleocene volcanic sequence north of Lake Clark agrees well with previous paleomagnetic investigations, cited in the Introduction, indicating no major latitudinal displacement of Upper Cretaceous to Eocene volcanic rocks from regions inboard of the Peninsular terrane. Moreover, our results indicates significant counterclockwise rotation. The gross structural grain of Alaska is consistent with such rotation and might be an expression of bending in response to convergence of Siberia and North America during Late Cretaceous and early Tertiary time (Grantz, 1966). If this great bend is indeed an orocline, the time of its formation has important bearing on the tectonic evolution of Alaska. The 45° to 55° counterclockwise rotation of southwest Alaska since the Late Cretaceous that is indicated with greater than 95% confidence by these and other (Globerman and Coe, 1984) paleomagnetic data, agrees well with the magnitude and timing of Eurasia–North America convergence inferred from magnetic anomalies (Coe et al., 1985). The lack of significant rotation of a middle Eocene sequence of flows north of Lake Clark and the 30–40 Ma Iliamna quadrangle localities is consistent with Eurasia–North America convergence terminating before the late Eocene.

The data south of Lake Clark show that an inboard part of the Peninsular terrane did not undergo significant northward translation after Eocene-Oligocene time. This evidence does not preclude the possibility that intra-Peninsular faulting facilitated substantial latitudinal displacement of outboard parts of the Peninsular terrane since the late Eocene. However, there is no indication of major northward displacement

Downloaded from geology.gsapubs.org on October 28, 2010

by new paleomagnetic data from Eocene-Oligocene rocks on Umnak Island of the Aleutian chain (Harbert and Cox, 1984) and on the Pacific coast of the Alaska Peninsula near Chignik (Thrupp and Coe, 1985; Whitney et al., 1985).

Outboard of the Peninsular terrane, paleomagnetic data from the Ghost Rocks volcanics on Kodiak Island indicate some 25° of northward displacement since the early Paleocene (Plumley et al., 1983). Moore et al. (1983) suggested that the Ghost Rocks paleomagnetic constraint applies to both the Prince William and Chugach terranes because the Paleocene rocks of the Gulf of Alaska magmatic belt intrude both terranes. On Kodiak Island, Davies and Moore (1984) found Paleocene dikes just inboard of the Border Ranges fault, which divides the Peninsular and Chugach terranes. They suggested that because the dikes appear to be appendages of the Gulf of Alaska magmatic belt, at least an outboard part of the Peninsular terrane may also have been part of the amalgam of exotic terranes since the Paleocene. Further evidence for close proximity of parts of the Chugach and Peninsular terranes in early Tertiary time is the occurrence of clasts of probable Chugach terrane rocks in the Paleocene Chickaloon Formation of the Peninsular terrane (Little, 1984). Moreover, in the northern Chugach Mountains, Eocene dikes span and locally intrude the Border Ranges fault (Little, 1984). An even earlier union (pre-Albian) of parts of these two terranes is indicated in the western Chugach Mountains by 100–125 Ma plutons that clearly crosscut the Border Ranges fault zone and other major faults with similar geometry nearby in the Peninsular terrane (Pavlis, 1982).

Paleomagnetic studies indicate that about 2000 km of latitudinal convergence occurred between the Ghost Rocks of Kodiak Island and more inboard Alaska since early Paleocene time. The geologic ties outlined above and the magmatic belt tie between the Alaska Peninsula and adjacent inboard undisplaced latest Cretaceous–early Eocene rocks would appear either to preclude such large-scale convergence or to restrict the locus of displacement to a narrow zone of the Peninsular terrane between its outboard margin and the Alaska–Aleutian Range. We suspect, however, that the magmatic ties are not so tight as generally interpreted and that large cumulative intraterrane displacements account for much of the northward movement. This seems plausible because (1) internal transcurrent displacement or thrusting may have significantly altered the original configuration and extent of a magmatic belt; (2) shearing would tend to skirt plutons and concentrate within weaker surrounding sediments; (3) igneous rocks used to define a magmatic belt may not all be cogenetic; and (4) if the evidence, discussed above, supporting oroclinal bending is correct, then because the margin of southern Alaska prior to middle Eocene time would have trended more northwesterly, dextral faulting would have facilitated northward translation.

Moreover, geologic evidence supporting major northward displacement during the Tertiary within the Cordilleran margin of northwest North America is difficult to reconcile without substantial intraterrane faulting. Cowan (1982) noted that occurrences of distinctive schists on Baranof and Vancouver islands, now separated by 1125 km, may have been contiguous only 40 m.y. ago. He speculated that such tectonic transport may have affected the entire Chugach terrane and a sliver of Wrangellia as well. Pavlis (1982) appealed to a similar amount of translation to explain the mismatch of the Chugach accretionary complex and an appropriate middle Cretaceous arc. Along the eastern Gulf of Alaska margin, plutons within Cretaceous flysch characteristically are sheared and altered (Hudson, 1983).

Numerous right-lateral strike-slip faults have been discovered in the Maastrichtian Kodiak Formation just inboard of the boundary between the Chugach and Prince William terranes (Sample, 1985). Detailed mapping of the Maastrichtian Valdez Formation on the Kenai Peninsula has also revealed abundant dextral strike-slip faults (Mitchell, 1979). This shearing within the Chugach terrane is commonly concentrated in weak, easily eroded, fine-grained sediments; consequently, surface expression of the faults is poor. Within a mine, however, Mitchell (1979) observed faults that occur locally every 10 cm and clearly cut early Eocene dikes presumably associated with the Gulf of Alaska magmatic belt. Although the amount of right-lateral offset, when determinable, on individual faults generally does not exceed a few metres, Mitchell speculated that 10 faults per metre may be representative of the region.

In the Prince William terrane, further suggestion of large intraterrane displacement is the discordance between the results from the two paleomagnetic sampling areas in the Ghost Rocks Formation of Kodiak Island. Plumley et al. (1983) suggested that the 122° difference between their two mean declinations represents large tectonic rotations associated with shearing in the Ghost Rocks. The inclination discrepancy is consistent with 15° of latitudinal separation of the two localities at 62 Ma, the more outboard locality lying farther south. Although it certainly seems plausible that the inclination discordance could be a consequence of secular variation as proposed by Plumley et al. (1983), appreciable dispersion of the flow means suggests that a fair amount of time may be represented at each locality. Thus, significant differential strike-slip displacement of Prince William terrane slivers may have occurred.

Considerable latitudinal displacement has probably been accommodated farther inboard as well. For instance, compression at the bend in the Denali fault that resulted in folding, thrusting, and splaying of strike-slip faults has absorbed a substantial part of the 400 km of translation that occurred during the Tertiary along the Denali fault system southeast of the bend (Nockleberg et al., 1985). Within the Peninsular terrane, post–early Eocene folding and reverse and strike-slip faulting are evident in the Chickaloon Formation (Little, 1984). Although not significant at high confidence levels, the paleolatitudes indicated by our paleomagnetic data are also consistent with modest early Tertiary northward displacement: 9° since the early Paleocene and less than 1° since the middle Eocene.

CONCLUSIONS AND SPECULATION

Our paleomagnetic results indicate that the Alaska Peninsula did not experience the gross northward displacement during the Tertiary

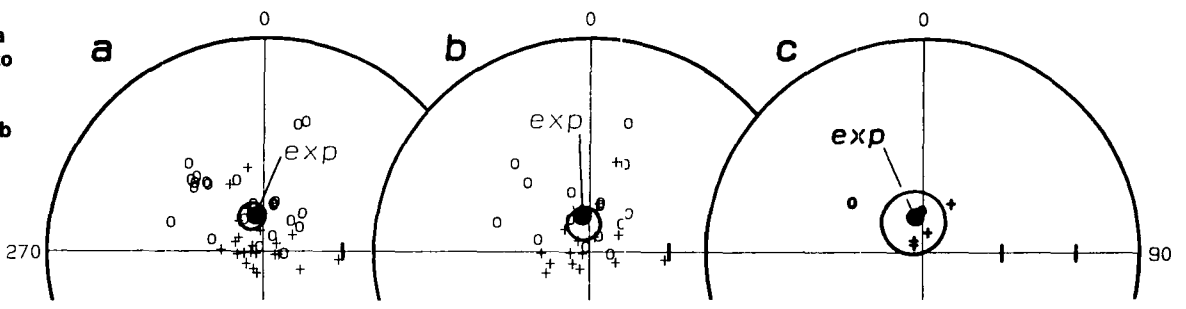


Figure 4. Peninsular terrane paleomagnetic data from 50 flows of age 30 to 40 Ma. a: Flow mean directions and their mean and alpha-95 (see Fig. 3b caption). b: Flow-group mean directions and alpha-95 about their mean. c: Flow-group locality means and alpha-95 about their mean.

Downloaded from geology.gsapubs.org on October 28, 2010

that has been detected in the Gulf of Alaska region, southeast Alaska, and British Columbia. Where and how to account for the major Tertiary northward translation of the outboard Gulf region relative to more inboard Alaska remains a matter of conjecture. However, it would not be surprising if terranes of this actively evolving continental margin at a plate boundary dominated by oblique convergence and transcurrent faulting did not remain coherent tectonic entities.

The magmatic belts in southern Alaska probably do not constitute rigorous ties. Significant Tertiary deformation affected the eastern part of the Peninsular terrane and Wrangellia (Nokleberg et al., 1985; Little, 1984). Detailed mapping has revealed strike-slip faults within the Chugach terrane that postdate Paleocene-Eocene Gulf of Alaska belt plutons (Mitchell, 1979; Sample, 1985). Although offset on individual faults appears small, cumulative displacement within the Chugach terrane may amount to hundreds of kilometres. This dextral faulting would have facilitated northward movement if the southern Alaska margin had a more northerly trend in the early Tertiary.

The Kodiak Island paleomagnetic data (Plumley et al., 1983) that indicate a large discrepancy in paleolatitude relative to stable North America also suggest the possibility that several hundred kilometres of northward translation were accommodated by dextral slivering within the Prince William terrane. Our results suggest that some additional displacement was taken up inboard of the Alaska Peninsula during Paleogene time. Experimental uncertainties in the paleomagnetic results could further lessen the residual discrepancy in latitude by at least a few hundred kilometres. The remaining northward movement of the outboard edge of Kodiak Island was probably accomplished by transcurrent faulting distributed within the Chugach and Peninsular terranes as well as shortening south of the Denali fault in the region of the great bend. Thus, it is likely that intraterrane displacements during the Tertiary played an important role in the postaccretionary reorganization of the Alaskan and Canadian margin.

REFERENCES CITED

Coe, R.S., Globerman, B.R., Plumley, P.W., and Thrupp, G.A., 1985, Paleomagnetic results from Alaska and their tectonic implications, *in* Howell, D.G., ed., Tectonostratigraphic terranes of the Circum-Pacific region: American Association of Petroleum Geologists Circum-Pacific Earth Sciences Series no. 1, p. 85–108.

Cowan, D.S., 1982, Geological evidence for post-40 m.y. B.P. large-scale northwestward displacement of part of southeastern Alaska: Geology, v. 10, p. 309–313.

Cox, A., and Gordon, R.G., 1984, Paleolatitudes determined from paleomagnetic data from vertical cores: Reviews of Geophysics and Space Physics, v. 22, p. 47–72.

Davies, D.L., and Moore, J.C., 1984, Sixty m.y. intrusive rocks from the Kodiak Islands link the Peninsular, Chugach, and Prince William terranes: Geological Society of America Abstracts with Programs, v. 16, p. 277.

Demarest, H.H., Jr., 1983, Error analysis for the determination of tectonic rotation from paleomagnetic data: Journal of Geophysical Research, v. 88, p. 4321–4328.

Detterman, R.L., and Reed, B.L., 1980, Stratigraphy, structure, and economic geology of the Iliamna quadrangle, Alaska: U.S. Geological Survey Bulletin 1368-B, 86 p.

Eakins, G.R., Gilbert, W.G., and Bundtzen, T.K., 1978, Preliminary bedrock geology and mineral resource potential of west-central Lake Clark quadrangle, Alaska: Alaska State Geological Survey Open-File Report 118, 13 p.

Globerman, B.R., and Coe, R.S., 1984, Discordant declinations from "in place" Upper Cretaceous volcanics, SW Alaska: Evidence for oroclinal bending?: Geological Society of America Abstracts with Programs, v. 16, p. 286.

Grantz, A., 1966, Strike-slip faults in Alaska: U.S. Geological Survey Open-File Report 267, 82 p.

Grommé, C.S., and Hillhouse, J.W., 1981, Paleomagnetic evidence for northward movement of the Chugach terrane, southern and southeastern Alaska: U.S. Geological Survey Circular 823-B, p. 161–164.

Harbert, W.P., and Cox, A., 1984, Preliminary paleomagnetic results from Umnak Island, Aleutian Ridge: Geological Society of America Abstracts with Programs, v. 16, p. 288.

Hillhouse, J.W., 1977, Paleomagnetism of the Triassic Nikolai Greenstone, McCarthy quadrangle, Alaska: Canadian Journal of Earth Sciences, v. 14, p. 2578–2592.

Hillhouse, J.W., and Grommé, C.S., 1982, Limits to northward drift of the Paleocene Cantwell Formation, central Alaska: Geology, v. 10, p. 552–556.

——1984, Northward displacement and accretion of Wrangellia: New paleomagnetic evidence from Alaska: Journal of Geophysical Research, v. 89, p. 4461–4477.

Hillhouse, J.W., Grommé, C.S., and Csejtey, B., Jr., 1985, Tectonic implications of paleomagnetic poles from early Tertiary volcanic rocks, south-central Alaska: Journal of Geophysical Research, v. 90.

Hudson, T., 1979, Mesozoic plutonic belts of southern Alaska: Geology, v. 7, p. 230–234.

——1983, Calc-alkaline plutonism along the Pacific rim of southern Alaska, *in* Roddick, J.A., ed., Circum-Pacific plutonic terranes: Geological Society of America Memoir 159, p. 159–169.

Jones, D.L., Silberling, N.J., Berg, H.C., and Plafker, G., 1981, Map showing tectono-stratigraphic terranes of Alaska, columnar sections, and summary description of terranes: U.S. Geological Survey Open-File Report 81-792, 20 p.

Little, T.A., 1984, Early Tertiary structural development of the northern Chugach Mountains, west of the Matanuska Glacier, AK: Geological Society of America Abstracts with Programs, v. 16, p. 294.

Mitchell, P.A., 1979, Geology of the Hope-Sunrise (Gold) Mining District, north central Kenai Peninsula, Alaska [M.S. thesis]: Stanford, California, Stanford University, 123 p.

Moore, J.C., Byrne, T., Plumley, P.W., Reid, M., Gibbons, H., and Coe, R.S., 1983, Paleogene evolution of the Kodiak Islands, Alaska: Consequences of ridge-trench interaction in a more southerly latitude: Tectonics, v. 2, p. 265–293.

Newton, C.R., 1983, Paleozoogeographic affinities of Norian bivalves from the Wrangellian, Peninsular, and Alexander terranes, western North America, *in* Stevens, C.H., ed., Pre-Jurassic rocks in western North American suspect terranes: Pacific Section, Society of Economic Paleontologists and Mineralogists, p. 37–48.

Nokleberg, W.J., Jones, D.L., and Silberling, N.J., 1985, Origin and tectonic evolution of the Maclaren and Wrangellia terranes, eastern Alaska Range, Alaska: Geological Society of America Bulletin, v. 96, p. 1251–1270.

Packer, D.R., and Stone, D.R., 1974, Paleomagnetism of Jurassic rocks from southern Alaska, and the tectonic implications: Canadian Journal of Earth Sciences, v. 11, p. 976–997.

Pavlis, T.L., 1982, Origin and age of the Border Ranges fault of southern Alaska and its bearing on the late Mesozoic evolution of Alaska: Tectonics, v. 1, p. 343–368.

Plumley, P.W., Coe, R.S., and Byrne, T., 1983, Paleomagnetism of the Paleocene Ghost Rocks Formation, Prince William Terrane, Alaska: Tectonics, v. 2, p. 295–314.

Sample, J., 1985, An ancient underplated sequence preserved in the Kodiak Islands, Alaska: Geological Society of America Abstracts with Programs, v. 17, p. 406.

Silberling, N.J., and Jones, D.L., 1984, Lithotectonic terrane maps of the North American Cordillera: U.S. Geological Survey Open-File Report 84-523, 12 p.

Stone, D.B., Panuska, B.C., and Packer, D.R., 1982, Paleolatitudes versus time for southern Alaska: Journal of Geophysical Research, v. 87, p. 3697–3707.

Thrupp, G.A., and Coe, R.S., 1985, Paleomagnetism of Eocene and Oligocene volcanic rocks on the Alaska Peninsula: Geological Society of America Abstracts with Programs, v. 17, p. 413.

Wallace, W.K., and Engebretson, D.C., 1984, Relationships between plate motions and Late Cretaceous to Paleocene magmatism in southwestern Alaska: Tectonics, v. 3, p. 295–315.

Whitney, J.W., Levinson, R.A., and Van Alstine, D.R., 1985, Paleomagnetism of early Tertiary Alaska Peninsula rocks and implications for docking of Peninsular terrane: Program and Abstracts of the Pacific Section Annual Meeting, American Association of Petroleum Geologists, Society of Economic Paleontologists and Mineralogists, Society of Exploration Geophysicists, p. 32.

ACKNOWLEDGMENTS

Supported by National Science Foundation Grants EAR-8206918 and EAR-8417639; ARCO Alaska Inc.; and Earth Sciences Board, University of California, Santa Cruz. S. Grommé suggested the Iliamna region. We are grateful for helicopter support by W. Wallace and J. Rogers of ARCO. B. Thrupp, A. Bol, and M. Garrison helped in the field. We thank P. Poe and the rest of the University of Washington Fisheries Research Institute Iliamna crew, the Hornbergers, Frita Slye, the Alsworths, Iliamna Air Taxi, and many other people of Iliamna and Lake Clark for their assistance, advice, and kindness. A review by J. Hillhouse and criticism of an early draft by S. Longiaru were helpful.

Manuscript received May 22, 1985
Revised manuscript received October 31, 1985
Manuscript accepted November 14, 1985